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Inventor(s)

Breault; Andrew E. et al.

CONTINUOUS FLOW FUEL SYSTEM

Abstract

A fuel system for a gas turbine engine includes a manifold assembly with an assembly inlet, an assembly outlet, and a plurality of fuel injectors. The manifold assembly also includes at least one fuel manifold fluidically connecting each fuel injector of the plurality of fuel injectors to the assembly inlet and to the assembly outlet. A fuel return line is fluidically connected to the assembly outlet. A fuel circulator pump includes a pump outlet fluidically connected to the assembly inlet.

Inventors: Breault; Andrew E. (Bolton, CT), Sobanski; Jon Erik (Glastonbury, CT), Morton; Jeffrey T. (Manchester, CT)

Applicant: RTX Corporation (Farmington, CT)

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Background/Summary

BACKGROUND

[0001] The present disclosure relates to gas turbine engines and, more particularly, to reducing coking in the fuel system.

[0002] Fuel nozzles inject fuel into a combustor of gas turbine engines. During operation, fuel flow through the fuel nozzles of the fuel system is used to thermally condition the nozzles and prevent coking from occurring in the nozzles and neighboring fuel lines. At shutdown of a gas turbine engine, fuel flow is halted and heat from the combustor and case structure soaks into the fuel system, leading to coking in the fuel system. Coke buildup compromises fuel nozzle performance and drives maintenance.

SUMMARY

[0003] In one example, a fuel system for a gas turbine engine includes a manifold assembly with an assembly inlet, an assembly outlet, and a plurality of fuel injectors. The manifold assembly also includes at least one fuel manifold fluidically connecting each fuel injector of the plurality of fuel injectors to the assembly inlet and to the assembly outlet. A fuel return line is fluidically connected to the assembly outlet. A fuel circulator pump includes a pump outlet fluidically connected to the assembly inlet.

[0004] In another example, a method is disclosed for cooling a fuel system of a gas turbine engine. The method includes shutting down the gas turbine engine. After shutting down the gas turbine engine, a fuel circulator pump is activated to flow fuel into at least one fuel manifold and a plurality of fuel injectors fluidically connected to the at least one fuel manifold. The fuel is directed from the at least one fuel manifold and the plurality of fuel injectors into a fuel return line.

[0005] In another example, a fuel injector for a gas turbine engine includes a strut and a nozzle head connected to the strut. The nozzle head includes a nozzle exit. A supply passage extends through the strut to the nozzle head. A return passage extends through the strut toward the nozzle head. A valve is within the strut and/or nozzle head and fluidically connects the supply passage to the nozzle exit.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0006] FIG. 1 is a partial cross-sectional view of a gas turbine engine.

[0007] FIG. 2 is a simplified partial cross-sectional view of a high pressure compressor section, combustor, and high pressure turbine section of the gas turbine engine of FIG. 1.

[0008] FIG. 3 is a schematic illustration of a fuel system with a fuel circulator pump and a return line for circulating fuel through an array of fuel injectors post shutdown of the gas turbine engine of FIG. 1.

[0009] FIG. 4 is a schematic illustration of another example of a fuel system with a fuel circulator pump and at least one return line for circulating fuel through an array of fuel injectors post shutdown of the gas turbine engine of FIG. 1.

[0010] FIG. 5 is a schematic illustration of another example of a fuel system with a fuel circulator pump and a return line for circulating fuel through an array of fuel injectors post shutdown of the gas turbine engine of FIG. 1.

[0011] FIG. 6 is a schematic illustration of another example of a fuel system with a fuel circulator pump and a return line for circulating fuel through an array of fuel injectors post shutdown of the gas turbine engine of FIG. 1.

[0012] FIG. 7 is a schematic illustration of another example of a fuel system with a fuel circulator pump and a return line for circulating fuel through an array of fuel injectors post shutdown of the gas turbine engine of FIG. 1.

[0013] FIG. 8 is a schematic illustration of another example of a fuel system with an electric fuel pump and a return line for circulating fuel through an array of fuel injectors post shutdown of the gas turbine engine of FIG. 1.

[0014] FIG. 9A is a schematic illustration of a fuel injector of the fuel system of any of FIGS. 3-8

with an actuator, a valve in an “on” position, and a return line.

[0015] FIG. **9B** is a cross-sectional view of a strut of the fuel injector of FIG. **9A** taken along line A-A.

[0016] FIG. **9C** is a schematic illustration of the fuel injector of FIG. **9A** with the valve in an “off” position.

[0017] FIG. **10A** is a schematic illustration of a fuel injector of the fuel system of any of FIGS. **3-8** with an internal valve in a closed position, a return line, and a shut-off valve in an open position.

[0018] FIG. **10B** is a cross-sectional view of a strut of the fuel injector of FIG. **10A** taken along line B-B.

[0019] FIG. **10C** is a schematic illustration of the fuel injector of FIG. **10A** with the internal valve in an open position and the shut-off valve in a closed position.

[0020] FIG. **11** is a schematic illustration of another example of a fuel system with a fuel circulator pump and a circulation circuit for circulating fuel through an array of fuel injectors post shutdown of the gas turbine engine of FIG. **1**.

[0021] While the above-identified drawing figures set forth one or more embodiments of the invention, other embodiments are also contemplated. In all cases, this disclosure presents the invention by way of representation and not limitation. It should be understood that numerous other modifications and embodiments can be devised by those skilled in the art, which fall within the scope and spirit of the principles of the invention. The figures may not be drawn to scale, and applications and embodiments of the present invention may include features and components not specifically shown in the drawings. Like reference numerals identify similar structural elements.

DETAILED DESCRIPTION

[0022] The disclosure relates to a fuel system of a gas turbine engine with a manifold and a fuel circulator pump that circulates fuel through the fuel system when the gas turbine engine is off. Fuel injectors of the fuel system include internal valving that prevents the fuel injectors from injecting fuel into a combustor of the gas turbine engine when the gas turbine engine is off. The internal valving can also allow the fuel to circulate through the fuel injectors while the fuel circulates through the fuel system. Allowing fuel to circulate through the fuel system and the fuel injectors during and/or after shut down of the gas turbine engine mitigates the formation of local fuel hotspots resulting from increased levels of conduction and/or radiation from the gas turbine engine. Avoiding local hot spots will reduce the likelihood of coke forming in the fuel injectors and the lines of the fuel system as heat soaks back through the cases of the gas turbine engine and into the fuel system. The reduced likelihood of coke formation will maintain fuel injector performance by maintaining intended fuel flow splits and distribution, which in turn maintains ideal combustor pattern factor. Maintaining fuel injector performance results in improved reliability and time on wing of the gas turbine engine. The fuel system is discussed below with reference to the figures.

[0023] FIG. **1** is a quarter-sectional view that schematically illustrates example gas turbine engine **20** that includes fan section **22**, compressor section **24**, combustor section **26** and turbine section **28**. Alternative engines might include an augmentor section (not shown) among other systems or features. Fan section **22** drives air along bypass flowpath BF while compressor section **24** draws air in along core flowpath C where air is compressed and communicated to combustor section **26**. In combustor section **26**, air is mixed with fuel and ignited to generate a high pressure exhaust gas stream that expands through turbine section **28** where energy is extracted and utilized to drive fan section **22** and compressor section **24**. Core casing **25** is disposed around compressor section **24**, combustor section **26** and turbine section **28** and separates these sections from bypass flowpath BF.

[0024] Although the disclosed non-limiting embodiment depicts a turbofan gas turbine engine, it should be understood that the concepts described herein are not limited to use with turbofans as the teachings may be applied to other types of turbine engines; for example, an industrial gas turbine; an open rotor gas turbine engine; a reverse-flow gas turbine engine; and a turbine engine including a three-spool architecture in which three spools concentrically rotate about a common axis CA

where a low spool enables a low pressure turbine to drive a fan or open rotor via a gearbox, an intermediate spool that enables an intermediate pressure turbine to drive a first compressor of the compressor section, and a high spool that enables a high pressure turbine to drive a high pressure compressor of the compressor section.

[0025] The example gas turbine engine **20** generally includes low-speed spool **30** and high-speed spool **32** mounted for rotation about center axis CA of gas turbine engine **20** relative to engine static structure **36** via several bearing systems **38**. It should be understood that various bearing systems **38** at various locations may alternatively or additionally be provided.

[0026] Low-speed spool **30** generally includes inner shaft **40** that connects fan **42** and low pressure (or first) compressor section **44** to low pressure (or first) turbine section **46**. Inner shaft **40** drives fan **42** through a speed change device, such as geared architecture **48**, to drive fan **42** at a lower speed than low-speed spool **30**. High-speed spool **32** includes outer shaft **50** that interconnects high pressure (or second) compressor section **52** and high pressure (or second) turbine section **54**. Inner shaft **40** and outer shaft **50** are concentric and rotate via bearing systems **38** about center axis CA.

[0027] Combustor **56** is arranged between high pressure compressor **52** and high pressure turbine section **54**. In one example, high pressure turbine section **54** includes at least two stages to provide double stage high pressure turbine section **54**. In another example, high pressure turbine section **54** includes only a single stage. As used herein, a “high pressure” compressor or turbine experiences a higher pressure than a corresponding “low pressure” compressor or turbine.

[0028] Mid-turbine frame **58** of engine static structure **36** can be arranged generally between high pressure turbine section **54** and low pressure turbine **46**. Mid-turbine frame **58** further supports bearing systems **38** in turbine section **28** as well as setting airflow entering the low pressure turbine **46**.

[0029] The gas flow in core flowpath C is compressed first by low pressure compressor **44** and then by high pressure compressor **52** mixed with fuel and ignited in combustor **56** to produce high pressure exhaust gases that are then expanded through high pressure turbine section **54** and low pressure turbine **46**. Mid-turbine frame **58** includes vanes **60**, which are in the core flowpath and function as an inlet guide vane for low pressure turbine **46**.

[0030] FIG. 2 is a simplified partial cross-sectional view of a high pressure compressor **52**, combustor **56**, and high pressure turbine section **54** of gas turbine engine **20** of FIG. 1. In the example of FIG. 2, gas turbine engine **20** further includes diffuser **62**, diffuser case **64**, and fuel system **66**. Fuel system **66** includes a plurality of fuel injectors **68** (only one of which is shown in FIG. 2) and combustor **56** include combustion chamber **70**.

[0031] As discussed above, combustor **56** is positioned in core flowpath C between high pressure compressor **52** and high pressure turbine section **54**. Diffuser case **64** can be a portion of core casing **25** that extends around diffuser **62**. Diffuser **62** is disposed radially within diffuser case **64** relative to center axis CA and is immediately upstream from combustor **56** relative to the gas flow in core flowpath C. Diffuser **62** is configured to slow the velocity of the gas flow in core flowpath C before the gas flow enters combustion chamber **70**. Each of fuel injectors **68** extends through diffuser case **64** and includes a nozzle that delivers fuel into combustion chamber **70**.

[0032] During operation of gas turbine engine **20**, the gas flow in core flowpath is compressed by low pressure compressor **44** (shown in FIG. 1) and by high pressure compressor **52**. After the gas flow in core flowpath C is compressed by high pressure compressor **52**, the gas flow then enters diffuser **62** where the gas flow decreases in velocity prior to entering combustion chamber **70** of combustor **56**. Upon entering combustion chamber **70**, the gas flow is mixed with fuel injected into combustion chamber **70** by fuel injectors **68**. The gas flow and fuel mixture is then ignited in combustion chamber **70** to produce high pressure exhaust gases that are then expanded through high pressure turbine section **54** and low pressure turbine **46** (shown in FIG. 1). As the fuel flows to and through fuel injectors **68** during operation of gas turbine engine **20**, the flow of the fuel helps cool fuel system **66** and prevent coking of the fuel in fuel system **66**.

[0033] After shutdown of gas turbine engine 20, residual heat in combustor 56, core casing 25, high pressure compressor 52 and high pressure turbine section 54 has the potential to cause heat soak-back in gas turbine engine 20. Heat soak-back occurs in gas turbine engines after shutdown when heat accumulated in hot parts and cavities of a gas turbine engine is transmitted to cooler parts. To protect fuel system 66 from heat soak-back after shutdown of gas turbine engine 20, fuel system 66 can continue to flow and circulate fuel through fuel injectors 68 without injecting the fuel into combustion chamber 70 to cool fuel system 66, as discussed below with reference to FIGS. 3-11.

[0034] FIG. 3 is a schematic illustration of an example of fuel system 66 with fuel circulator pump 72. In addition to fuel circulator pump 72, fuel system 66 can also include manifold assembly 74, fuel supply line 76, fuel return line 78, main fuel pump 80, fuel tank 82, main pump line 84, and check valve 86. In the example of FIG. 3, fuel circulator pump 72 includes pump outlet 88, pump inlet 90, and shut off valve 92. Manifold assembly 74 includes assembly inlet 94, assembly outlet 96, fuel injectors 68, and fuel manifolds 98. Each fuel injector 68 includes strut 100, nozzle head 102, and nozzle exit 106. Main fuel pump 80 includes main pump inlet 108 and main pump outlet 110.

[0035] Fuel injectors 68 are arrayed and circumferentially spaced apart from one another about center axis CA (shown in FIGS. 1 and 2) of gas turbine engine 20. Fuel manifolds 98 each include tubing and fittings that allow fuel manifolds 98 to fluidically connect to plumbing (not shown) of struts 100 of fuel injectors 68. Fuel manifolds 98 fluidically connect to fuel injectors 68 such that fuel manifolds 98 and fuel injectors 68 form a flow path that fluidically connects assembly inlet 94 to assembly outlet 96. Nozzle head 102 of each fuel injector 68 is connected to strut 100 for physical support and to receive fuel from the plumbing of strut 100. Nozzle exit 106 is formed on an end of nozzle head 102 so that fuel can be sprayed from nozzle head 102 into combustion chamber 70 (shown in FIG. 2). Internal valve 104 is inside strut 100 and/or nozzle head 102 and is configured to fluidically connect and disconnect nozzle exit 106 from the plumbing of strut 100.

[0036] Fuel supply line 76 fluidically connects pump outlet 88 of fuel circulator pump 72 to assembly inlet 94 of manifold assembly 74. In the example of FIG. 3, fuel return line 78 fluidically connects assembly outlet 96 of manifold assembly 74 to pump inlet 90 of fuel circulator pump 72. Shut off valve 92 can be housed within fuel circulator pump 72 or can be fluidically adjacent to fuel circulator pump 72 (i.e., immediately upstream or downstream from fuel circulator pump 72). Shut off valve 92 is configured to open and close pump outlet 88 and/or pump inlet 90. Shut off valve 92 is open when fuel circulator pump 72 is powered and active. Shut off valve 92 is closed when fuel circulator pump 72 is inactive. Activation, deactivation, and control of fuel circulator pump 72 and shut off valve 92 can be controlled by an electronic engine control (EEC), which is not shown. Fuel circulator pump 72 can be powered by an onboard battery bank or some other power source that does not require gas turbine engine 20 to be powered up.

[0037] Fuel tank 82 is fluidically connected to main pump inlet 108. Main pump line 84 is a fuel line that provides a fluid pathway from main pump outlet 110 of main fuel pump 80 to assembly inlet 94. Check valve 86 is connected to main pump line 84 and is positioned fluidically between main pump outlet 110 and assembly inlet 94. When check valve 86 is open, assembly inlet 94 is fluidically connected to main pump outlet 110 and can receive fuel from main fuel pump 80. Check valve 86 opens when main fuel pump 80 is active and pumping fuel from fuel tank 82 to manifold assembly 74. Check valve 86 is closed when main fuel pump 80 is inactive and while fuel circulator pump 72 is powered and actively circulating fuel through manifold assembly 74. Activation, deactivation, and control of main fuel pump 80 can be controlled by the electronic engine control (EEC), which is not shown.

[0038] In the example of FIG. 3, when gas turbine engine 20 is in operation, main fuel pump 80 is active and supplying fuel to manifold assembly 74 while fuel circulator pump 72 is inactive and shut off valve 92 is closed. Main fuel pump 80 pulls fuel from fuel tank 82 into main pump inlet 108 and pushes the fuel through main pump outlet 110 into main pump line 84. The pressure of the

fuel flowing in main pump line **84** from main fuel pump **80** opens check valve **86**. The fuel passes through check valve **86** and continues through main pump line **84** to assembly inlet **94** of manifold assembly **74**. The fuel enters manifold assembly **74** through assembly inlet **94** and flows through fuel manifolds **98** to each fuel injector **68**. The fuel enters each fuel injector **68** through the plumbing of strut **100**. During operation of gas turbine engine **20**, internal valve **104** is open such that the fuel can enter nozzle head **102** from strut **100** and can pass through nozzle exit **106** into combustion chamber **70** of combustor **56** (shown in FIG. **2**) to be mixed with air and ignited.

[0039] While gas turbine engine **20** and main fuel pump **80** are active, fuel circulator pump **72** is inactive and shut off valve **92** is closed to prevent fuel from flowing through fuel circulator pump **72**. During active operation of gas turbine engine **20** and main fuel pump **80**, flow of the fuel into manifold assembly **74** and out of nozzle exits **106** cools manifold assembly **74** sufficiently to prevent the fuel inside of manifold assembly **74** from coking and clogging fuel system **66**.

[0040] Immediately after gas turbine engine **20** shuts down, main fuel pump **80** is deactivated, internal valves **104** of fuel injectors **68** are closed, shut off valve **92** opens, and fuel circulator pump **72** is activated. Fuel circulator pump **72** circulates the fuel in manifold assembly **74** immediately after shutdown of gas turbine engine **20** to help cool manifold assembly **74** and prevent coking within fuel system **66**. Fuel circulator pump **72** circulates the fuel in manifold assembly **74** by flowing fuel from assembly outlet **96**, through pump inlet **90**, through shut off valve **92**, through pump outlet **88**, through fuel supply line **76**, and into assembly inlet **94**. After passing through assembly inlet **94**, fuel circulator pump **72** causes the fuel to flow through each of fuel manifolds **98** and each of fuel injectors **68**. As the fuel flows through each of fuel injectors **68**, the fuel flows radially inward (relative to center axis CA shown in FIGS. **1** and **2**) through strut **100** to nozzle head **102**. Since internal valve **104** is closed, the fuel does not flow out of nozzle head **102** through nozzle exit **106**. Rather, the fuel makes a turn within fuel injector **68** and flows radially outward from nozzle head **102** through strut **100** to one of fuel manifolds **98**.

[0041] After flowing through fuel manifolds **98** and fuel injectors **68**, the fuel exits manifold assembly **74** through assembly outlet **96**. From assembly outlet **96**, the fuel passes through fuel return line **78** to return to pump inlet **90** of fuel circulator pump **72** to begin another pass through the fuel circuit formed by fuel circulator pump **72**, fuel supply line **76**, manifold assembly **74**, and fuel return line **78**. While fuel circulator pump **72** is active and main fuel pump **80** is inactive, the pressure of the fuel generated by fuel circulator pump **72** causes check valve **86** to close. Closure of check valve **86** prevents the fuel circulated by fuel circulator pump **72** from back flowing through main fuel pump **80** and escaping the fuel circuit formed by fuel circulator pump **72**, fuel supply line **76**, manifold assembly **74**, and fuel return line **78**.

[0042] Fuel circulator pump **72** continues to circulate the fuel through manifold assembly **74** until gas turbine engine **20** has sufficiently cooled down to a temperature that is too low to cause coking in fuel system **66**. At this point where gas turbine engine **20** has sufficiently cooled and there is insufficient heat in gas turbine engine **20** to cause coking in fuel system **66**, fuel circulator pump **72** can be deactivated and shut off valve **92** closed. Alternatively, fuel circulator pump **72** may circulate fuel through manifold assembly **74** on an intermittent basis until gas turbine **20** has sufficiently cooled down to a temperature that is too low to cause coking in fuel system **66**. Fuel circulator pump **72** may be scheduled to run for a set duration and a set interval or may be scheduled to reduce the duty cycle as gas turbine **20** cools and the risk of fuel coking in fuel system **66** decreases. Internal valves **104** in fuel injectors **68** can remain closed to prevent fuel from leaking out of fuel injectors **68** until the next startup of gas turbine engine **20**. Protecting fuel system **66** against coking helps keep passages, fuel filters, fuel hydraulics, and heat exchangers within fuel system **66** clear so that combustor **56** receives a steady supply of fuel during engine operation. Reducing or eliminating coking in fuel system **66** also reduces maintenance time and cost on fuel system **66** and gas turbine engine **20**. Fuel circulator pump **72** can also be activated intermittently during operation of gas turbine engine **20** to circulate fuel through manifold assembly **74** to prevent

coking within fuel supply line **76** and fuel return line **78**.

[0043] FIG. **4** is a schematic illustration of another example of fuel system **66** with first manifold assembly **74a** and second manifold assembly **74b** fluidically connected in parallel with fuel circulator pump **72**. In addition to fuel circulator pump **72**, first manifold assembly **74a**, and second manifold assembly **74b**, the example of fuel system **66** shown in FIG. **4** can also include fuel supply line **76**, first fuel return line **78a**, second fuel return line **78b**, main fuel pump **80**, fuel tank **82**, main pump line **84**, and check valve **86**. Fuel circulator pump **72** includes pump outlet **88**, pump inlet **90**, and shut off valve **92**. First manifold assembly **74a** includes assembly inlet **94a**, assembly outlet **96a**, fuel injectors **68a**, and fuel manifolds **98a**. Second manifold assembly **74b** includes assembly inlet **94b**, assembly outlet **96b**, fuel injectors **68b**, and fuel manifolds **98b**. Each of fuel injectors **68a** and fuel injectors **68b** includes strut **100**, nozzle head **102**, and nozzle exit **106**. Main fuel pump **80** includes main pump inlet **108** and main pump outlet **110**. The example of fuel system **66** of FIG. **4** is similar to the example of fuel system **66** of FIG. **3**. Thus, components of the example of fuel system **66** shown in FIG. **4** and components of the example of fuel system **66** shown in FIG. **3** that share like reference numerals also share similar structure and function.

[0044] In the example of FIG. **4**, fuel circulator pump **72**, fuel supply line **76**, first manifold assembly **74a**, second manifold assembly **74b**, first fuel return line **78a**, and second fuel return line **78b** form a closed flow circuit when main fuel pump **80** is inactive and fuel circulator pump **72** is active. First manifold assembly **74a** forms a first leg of the closed flow circuit and second manifold assembly **74b** forms a second leg of the closed flow circuit that is fluidically in parallel to first manifold assembly **74a**.

[0045] Fuel injectors **68a** and fuel injectors **68b** are arrayed and circumferentially spaced apart from one another about center axis CA (shown in FIGS. **1** and **2**) of gas turbine engine **20**. Both fuel injectors **68a** and fuel injectors **68b** shown in FIG. **4** are similar to fuel injectors **68** described above with reference to FIG. **3**. Thus, the description above of fuel injectors **68** also applies to fuel injectors **68a** and fuel injectors **68b**. Fuel manifolds **98a** fluidically connect to fuel injectors **68a** such that fuel manifolds **98a** and fuel injectors **68a** form a flow path that fluidically connects assembly inlet **94a** to assembly outlet **96a**. Fuel manifolds **98b** fluidically connect to fuel injectors **68b** such that fuel manifolds **98b** and fuel injectors **68b** form a flow path that fluidically connects assembly inlet **94b** to assembly outlet **96b**.

[0046] Fuel supply line **76** fluidically connects pump outlet **88** of fuel circulator pump **72** to assembly inlet **94a** of first manifold assembly **74a** and to assembly inlet **94b** of second manifold assembly **74b**. In the example of FIG. **4**, first fuel return line **78a** fluidically connects assembly outlet **96a** of first manifold assembly **74a** to pump inlet **90** of fuel circulator pump **72**. Second fuel return line **78b** fluidically connects assembly outlet **96b** of second manifold assembly **74b** to pump inlet **90** of fuel circulator pump **72**.

[0047] Main pump line **84** is a fuel line that provides a fluid pathway from main pump outlet **110** of main fuel pump **80** to both assembly inlet **94a** of first manifold assembly **74a** and assembly inlet **94b** of second manifold assembly **74b**. Check valve **86** is connected to main pump line **84** and is positioned fluidically between main pump outlet **110** and both assembly inlet **94a** and assembly inlet **94b**. When check valve **86** is open, assembly inlets **94a** and **94b** are fluidically connected to main pump outlet **110** and can receive fuel from main fuel pump **80**. Check valve **86** opens when main fuel pump **80** is active and pumping fuel from fuel tank **82** to first manifold assembly **74a** and second manifold assembly **74b**.

[0048] In the example of FIG. **4**, when gas turbine engine **20** is in operation, main fuel pump **80** is active and supplying fuel to first manifold assembly **74a** and second manifold assembly **74b** while fuel circulator pump **72** is inactive and shut off valve **92** is closed. Main fuel pump **80** pulls fuel from fuel tank **82** into main pump inlet **108** and pushes the fuel through main pump outlet **110** into main pump line **84**. The pressure of the fuel flowing in main pump line **84** from main fuel pump **80** opens check valve **86**. The fuel passes through check valve **86** and continues through main pump

line **84**. A first portion of the fuel in main pump line **84** is directed to assembly inlet **94a** of first manifold assembly **74a**. A second portion of the fuel in main pump line **84** is directed to assembly inlet **94b** of second manifold assembly **74b**.

[0049] The first portion of the fuel enters first manifold assembly **74a** through assembly inlet **94a** and flows through fuel manifolds **98a** to each fuel injector **68a**. The first portion of the fuel enters each fuel injector **68a** through the plumbing of strut **100**. During operation of gas turbine engine **20**, internal valve **104** is open such that the fuel can enter nozzle head **102** from strut **100** and can pass through nozzle exit **106** into combustion chamber **70** of combustor **56** (shown in FIG. 2) to be mixed with air and ignited. In similar fashion, the second portion of the fuel enters second manifold assembly **74b** through assembly inlet **94b** and flows through fuel manifolds **98b** to each fuel injector **68b**. The second portion of the fuel enters each fuel injector **68b** through the plumbing of strut **100**. During operation of gas turbine engine **20**, internal valve **104** is open such that the fuel can enter nozzle head **102** from strut **100** and can pass through nozzle exit **106** into combustion chamber **70** of combustor **56** (shown in FIG. 2) to be mixed with air and ignited.

[0050] While gas turbine engine **20** and main fuel pump **80** are active, fuel circulator pump **72** is inactive and shut off valve **92** is closed to prevent fuel from flowing through fuel circulator pump **72**. During active operation of gas turbine engine **20** and main fuel pump **80**, flow of the first portion of the fuel into first manifold assembly **74a** and out of nozzle exits **106** of fuel injectors **68a** cools first manifold assembly **74a**. Flow of the second portion of the fuel into second manifold assembly **74b** and out of nozzle exits **106** of fuel injectors **68b** cools second manifold assembly **74b**.

[0051] Immediately after gas turbine engine **20** shuts down, main fuel pump **80** is deactivated, internal valves **104** of fuel injectors **68a** and internal valves **104** of fuel injectors **68b** are closed, shut off valve **92** opens, and fuel circulator pump **72** is activated. Fuel circulator pump **72** circulates the fuel in first manifold assembly **74a** and second manifold assembly **74b** immediately after shutdown of gas turbine engine **20** to help cool first manifold assembly **74a** and cool second manifold assembly **74b** and prevent coking within fuel system **66**. Fuel circulator pump **72** circulates the fuel in first manifold assembly **74a** and second manifold **74b** by flowing fuel from both assembly outlet **96a** of first manifold assembly **74a** and assembly outlet **96b** of second manifold **74b** through pump inlet **90**. The fuel then passes through shut off valve **92**, through pump outlet **88**, and through fuel supply line **76**. The fuel then passes from fuel supply line **76** to assembly inlet **94a** of first manifold assembly **74a** and assembly inlet **94b** of second manifold assembly **74b**. The fuel is then divided between first manifold assembly **74a** and second manifold assembly **74b**. The fuel passes through first manifold assembly **74a** and second manifold assembly **74b** in similar fashion as manifold assembly **74** described above with reference to FIG. 3.

[0052] After flowing through first manifold assembly **74a** and second manifold assembly **74b**, the fuel exits first manifold assembly **74a** through assembly outlet **96a** and exits second manifold assembly **74b** through assembly outlet **96b**. From assembly outlets **96a** and **96b**, the fuel passes through first fuel return line **78a** and second fuel return line **78b** to return to pump inlet **90** of fuel circulator pump **72** to begin another pass through the fuel circuit formed by fuel circulator pump **72**, fuel supply line **76**, first manifold assembly **74a**, second manifold assembly **74b**, first fuel return line **78a**, and second fuel return line **78b**. While fuel circulator pump **72** is active and main fuel pump **80** is inactive, the pressure of the fuel generated by fuel circulator pump **72** causes check valve **86** to close. Closure of check valve **86** prevents the fuel circulated by fuel circulator pump **72** from back flowing through main fuel pump **80** and escaping the fuel circuit formed by fuel circulator pump **72**, fuel supply line **76**, first manifold assembly **74a**, second manifold assembly **74b**, first fuel return line **78a**, and second fuel return line **78b**.

[0053] Fuel circulator pump **72** continues to circulate the fuel through first manifold assembly **74a** and second manifold assembly **74b** until gas turbine engine **20** has sufficiently cooled down to a temperature that is too low to cause coking in fuel system **66**. At this point where gas turbine

engine **20** has sufficiently cooled and there is insufficient heat in gas turbine engine **20** to cause coking in fuel system **66**, fuel circulator pump **72** can be deactivated and shut off valve **92** closed. Internal valves **104** in fuel injectors **68a** and **68b** can remain closed to prevent fuel from leaking out of fuel injectors **68a** and **68b** until the next startup of gas turbine engine **20**. Protecting fuel system **66** against coking helps keep passages within fuel system **66** clear so that combustor **56** receives a steady and reliable supply of fuel during engine operation. Reducing or eliminating coking in fuel system **66** also reduces maintenance time and cost on fuel system **66** and gas turbine engine **20**, which in turn increases the operational availability of gas turbine engine **20**.

[0054] FIG. **5** is a schematic illustration of another example fuel system **66**. Fuel system **66** of FIG. **5** is similar to the example of FIG. **3**, with the addition of core turning motor **112** and coupling **114**. Core turning motor **112** is a motor that rotates inner shaft **40** and/or outer shaft **50** (both shown in FIG. **1**) after gas turbine engine **20** is powered down to mitigate bowing of inner shaft **40** and outer shaft **50**. Core turning motor **112** and fuel circulator pump **72** can be mechanically connected to each other and connected to low-speed spool **30** or high-speed spool **32**, through coupling **114**. In the example of FIG. **5**, coupling **114** can include a geartrain, clutch, and/or mechanical solenoid. Coupling **114** disengages core turning motor **112** and fuel circulator pump **72** from high speed spool **32** when gas turbine engine **20** is running. When gas turbine engine **20** is shut down and core turning motor **112** and fuel circulator pump **72** is activated, coupling **114** rotationally engages fuel circulator pump **72** with low-speed spool **30** or high-speed spool **32**. Thus, activation and powering of fuel circulator pump **72** can be dependent on activation and powering of core turning motor **112**. Core turning motor **112** can be housed on gas turbine engine **20**, or core turning motor **112** can be part of a ground-based servicing unit that connects to coupling **114** once the aircraft powered by gas turbine engine **20** has landed and parked.

[0055] FIG. **6** is a schematic illustration of another example fuel system **66**. Fuel system **66** of FIG. **6** is similar to the example of FIG. **3**, except a single fuel manifold **98** connects all of fuel injectors **68** to fuel system **66**. Fuel manifold **98** in FIG. **6** includes multiple jumper feeds **115** that fluidically connect fuel manifold **98** to fuel injectors **68**. In some examples, jumper feeds **115** can include both a supply passage and a return passage such that fuel can circulate through each fuel injector **68** when main fuel pump **80** is deactivated, fuel circulator pump **72** is activated, shut off valve **92** is open, and internal valves **104** are closed.

[0056] FIG. **7** is a schematic illustration of another example fuel system **66**. Fuel system **66** of FIG. **7** is similar to the example of FIG. **3**, except pump inlet **90** of fuel circulator pump **72** is fluidically connected to fuel tank **82**, fuel return line **78** fluidically connects assembly outlet **96** of manifold assembly **74** to fuel tank **82**, and fuel system **66** also includes return loop **116**. Return loop **116** can redirect excess fuel from main fuel pump **80** to fuel tank **82** during normal operation. In the example of FIG. **7**, fuel circulator pump **72** is circulating fuel from fuel tank **82** through manifold assembly **74**, through fuel return line **78** and back to fuel tank **82**. By moving fuel through fuel return line **78**, fuel circulator pump **90** can help manage the temperature of the fuel in manifold assembly **74** and protect fuel system **66** from coking.

[0057] FIG. **8** is a schematic illustration of another example of fuel system **66**. In the example of FIG. **8**, fuel circulator pump **72** is an electric fuel pump that also fulfills the function of main fuel pump **80** from FIG. **3**. Fuel circulator pump **72** can include first pump inlet **90a** fluidically connected fuel return line **78**, and second pump inlet **90b** fluidically connected to fuel tank **82**. Fuel circulator pump **72** can also include a selector valve (not shown) that can selectively open and close first pump inlet **90a** and second pump inlet **90b**.

[0058] In the example of FIG. **8**, during operation of gas turbine engine **20**, fuel circulator pump **72** is powered and moves fuel from fuel tank **82** through second pump inlet **90b**, through pump outlet **88**, and into fuel supply line **76**. The fuel travels in fuel supply line **76** to assembly inlet **94** and enters manifold assembly **74**. The fuel travels through fuel manifolds **98** to fuel injectors **68**. The fuel enters struts **100** of fuel injectors **68**, passes through internal valves **104** (which are open

during operation of gas turbine engine **20**), enters nozzle head **102**, and is sprayed from nozzle exit **106** into combustion chamber **70** (shown in FIG. 2). The fuel is mixed with air in combustion chamber **70** and ignited to produce hot exhaust gases. During operation of gas turbine engine **20**, the selector valve of fuel circulator pump **72** can close first pump inlet **90a** to prevent fuel circulator pump **72** from drawing fuel from manifold assembly **74** through assembly outlet **96** and fuel return line **78**.

[0059] When gas turbine engine **20** shuts down, internal valves **104** of fuel injectors **68** close to stop fuel from exiting out of fuel injectors **68** through nozzle exits **106**. The selector valve of fuel circulator pump **72** can close second pump inlet **90b** and open first pump inlet **90a** so that fuel circulator pump **72** can draw fuel from assembly outlet **96** through fuel return line **78**. Fuel circulator pump **72** then moves the fuel through pump outlet **88**, through fuel supply line **76**, and back into manifold assembly **74**. Thus, fuel circulator pump **72**, fuel supply line **76**, manifold assembly **74**, and fuel return line **78** can form a closed loop for circulating the fuel. In a variation of the example of FIG. 8, fuel return line **78** can be fluidically connected to fuel tank **82** instead of first pump inlet **90a**, such that fuel circulator pump **72** can draw fuel from fuel tank **82** to circulate through manifold assembly **74** post engine shutdown, and the fuel can return to fuel tank **82** after passing through manifold assembly **74**.

[0060] FIGS. 9A-9C will be discussed concurrently. FIG. 9A is a schematic illustration of an example of fuel injector **68** of fuel system **66** with internal valve **104** in an “on” position. FIG. 9B is a cross-sectional view of strut **100** of fuel injector **68** of FIG. 9A taken along line A-A. FIG. 9C is a schematic illustration of fuel injector **68** from FIG. 9A with internal valve **104** in an “off” position. The example of fuel injector **68** shown in FIGS. 9A-9C can be used by any of the examples of fuel system **66** shown in FIGS. 3-8. In the example of FIGS. 9A-9C, fuel injector **68** further includes supply passage **118** and return passage **120**. Internal valve **104** can be an electrically actuated valve that includes actuator **122**, drive shaft **124**, valve head **126**, first valve seat **128**, and second valve seat **130**.

[0061] In the example of FIGS. 9A-9C, first valve seat **128** and second valve seat **130** are in strut **100** proximate to nozzle head **102**. In other examples, first valve seat **128** and second valve seat **130** can be in nozzle head **102**. Supply passage **118** is a flow passage that extends along strut **100** from a radially outer end of strut **100** (relative to center axis CA) to second valve seat **130**. Return passage **120** is a flow passage that extends along strut **100** from the radially outer end of strut **100** to first valve seat **128**. Fuel manifolds **98** (shown in FIGS. 3-8) can fluidically connect supply passage **118** to assembly inlet **94** of manifold assembly **74**. Fuel manifolds **98** can fluidically connect return passage **120** to assembly outlet **96** of manifold assembly **74** (also shown in FIGS. 3-8).

[0062] Valve head **126** is positioned between first valve seat **128** and second valve seat **130**. Actuator **122** is positioned outside of diffuser case **64** to protect actuator **122** from the heat within diffuser case **64** generated by combustor **56**. Drive shaft **124** extends from actuator **122** through strut **100** to valve head **126**. Drive shaft **124** connects valve head **126** to actuator **122** such that actuator **122** can move valve head **126** between first valve seat **128** and second valve seat **130**. Actuator **122** can be an electrical motor, such as servo motor, that is controlled by the electronic engine control (EEC) to move valve head **126** from first valve seat **128** to second valve seat **130** and vice versa via drive shaft **124**.

[0063] When gas turbine engine **20** is on, actuator **122** can position valve head **126** against first valve seat **128** to close first valve seat **128** and fluidically disconnect return passage **120** from supply passage **118**. With valve head **126** against first valve seat **128**, second valve seat **130** is open and supply passage **118** is fluidically connected to nozzle head **102** and nozzle exit **106**. With supply passage **118** fluidically connected to nozzle exit **106**, fuel can travel through strut **100**, through nozzle head **102**, and through nozzle exit **106** to be mixed with air in combustion chamber **70** (shown in FIG. 2) and ignited.

[0064] When gas turbine engine **20** is off, or is turning off, actuator **122** can position valve head **126** against second valve seat **130** to close second valve seat **130** and fluidically disconnect supply passage **118** from nozzle exit **106**. With valve head **126** against second valve seat **130**, first valve seat **128** is open and supply passage **118** is fluidically connected to return passage **120**. With supply passage **118** fluidically connected to return passage **120**, fuel circulator pump **72** (shown in FIGS. **3-8**) can cause the fuel to circulate through fuel injector **68** by passing the fuel radially downward through supply passage **118**, turning the fuel at internal valve **104**, and passing the fuel radially upward through return passage **120**. After passing through return passage **120**, the fuel continues through manifold assembly **74**, eventually exits manifold assembly **74** through assembly outlet **96**, and passes through fuel return line **78**, as discussed above with reference to FIGS. **3-8**. While FIGS. **9A-9C** show internal valve **104** as an electrically actuated valve, internal valve **104** can also be a pressure release valve, as discussed below with reference to FIGS. **10A-10C**.

[0065] FIGS. **10A-10C** will be discussed concurrently. FIG. **10A** is a schematic illustration of an example of fuel injector **68** of fuel system **66** with internal valve **104** in a closed position. FIG. **10B** is a cross-sectional view of strut **100** of fuel injector **68** of FIG. **10A** taken along line B-B. FIG. **10C** is a schematic illustration of fuel injector **68** from FIG. **10A** with internal valve **104** in an open position. The example of fuel injector **68** shown in FIGS. **10A-10C** can be used by any of the examples of fuel system **66** shown in FIGS. **3-8**. In the example of FIGS. **10A-10C**, internal valve **104** can be a pressure release valve that includes valve head **132**, spring **134**, piston **136**, and valve seat **138**.

[0066] Valve seat **138** is in strut **100** proximate to nozzle head **102** and includes an opening that fluidically connects supply passage **118** to nozzle head **102**. In other examples, valve seat **138** can be in nozzle head **102**. Supply passage **118** is a flow passage that extends along strut **100** from a radially outer end of strut **100** (relative to center axis CA) to valve seat **138**. Return passage **120** is a flow passage that extends along strut **100** from the radially outer end of strut **100** toward valve seat **138**. In the example of FIGS. **10A-10C**, return passage **120** fluidically connects with supply passage **118** just above internal valve **104** relative to center axis CA and forms a continuous passage with supply passage **118**. Fuel manifolds **98** (shown in FIGS. **3-8**) can fluidically connect supply passage **118** to assembly inlet **94** of manifold assembly **74**. Fuel manifolds **98** can fluidically connect return passage **120** to assembly outlet **96** of manifold assembly **74** (also shown in FIGS. **3-8**).

[0067] Piston **136** is positioned on a first side of valve seat **138** and valve head **132** is positioned on a second side of valve seat **138**. A stem or connector can extend through valve seat **138** to connect piston **136** to valve head **132**. Spring **134** is positioned between piston **136** and valve head **132** and biases valve head **132** to a closed position against valve seat **138**. When in the closed position valve head **132** closes valve seat **138** and blocks fuel from entering nozzle head **102** and passing through nozzle exit **106**.

[0068] FIG. **10C** shows fuel injector **68** when gas turbine engine **20** is on. When gas turbine engine **20** is on, shut off valve **92** of circulator pump **72** is closed such that fuel cannot exit through assembly outlet **96** (shown in FIGS. **3-8**). The pressure of the fuel entering fuel injector **68** and the back pressure caused by shut off valve **92** being closed pushes on piston **136** and overcomes the biasing force of spring **134** to compress spring **134** and push valve head **132** away from valve seat **138**. As the fuel pushes valve head **132** away from valve seat **138**, valve seat **138** opens to fluidically connect supply passage **118** to nozzle head **102** and nozzle exit **106**. With supply passage **118** fluidically connected to nozzle exit **106**, fuel can travel through strut **100**, through nozzle head **102**, and through nozzle exit **106** to be mixed with air in combustion chamber **70** (shown in FIG. **2**) and ignited.

[0069] FIG. **10A** shows fuel injector **68** when gas turbine engine **20** is off or turning off. When gas turbine engine **20** is off, or is turning off, shut off valve **92** of circulator pump **72** is opened such that fuel can exit manifold assembly **74** through assembly outlet **96** (shown in FIGS. **3-8**). With shut

off valve **92** open, the hydraulic pressure of the fuel within supply passage **118** and return passage **120** drops and the biasing force of spring **134** forces valve head **132** against valve seat **138** to close valve seat **138** and fluidically disconnect supply passage **118** and return passage **120** from nozzle exit **106**. With valve seat **138** closed and shut off valve **92** open, fuel circulator pump **72** (shown in FIGS. **3-8**) can cause the fuel to circulate through fuel injector **68** by passing the fuel radially downward through supply passage **118**, turning the fuel near internal valve **104**, and passing the fuel radially upward through return passage **120**. After passing through return passage **120**, the fuel continues through manifold assembly **74**, eventually exits manifold assembly **74** through assembly outlet **96**, and passes through fuel return line **78**, as discussed above with reference to FIGS. **3-8**. [0070] FIG. **11** is a schematic illustration of another example fuel system **66** with first manifold assembly **140** dedicated to circulating fuel through fuel system **66** and second manifold assembly **142** dedicated to delivering fuel to combustor **56**. As shown in FIG. **11**, first manifold assembly **140** can include first assembly inlet **144**, first assembly outlet **146**, fuel injectors **148**, and primary fuel manifolds **150**. Second manifold assembly **142** can include second assembly inlet **152**, fuel injectors **148**, and secondary fuel manifolds **154**. Each fuel injector **148** can include strut **156**, nozzle head **158**, nozzle exit **160**, first passage **162**, and second passage **164**.

[0071] First passage **162** in each fuel injector **148** is a cooling passage that extends radially inward, relative to center axis CA, through strut **156** to nozzle head **158** and then turns and extends radially outward through strut **156**. First passage **162** is solely a cooling passage in fuel injector **148** and does not fluidically connect with nozzle exit **160**. Second passage **164** in each fuel injector **148** is a fuel supply passage that extends radially inward, relative to center axis CA, through strut **156** to nozzle head **158** to fluidically connect with nozzle exit **160**. Second passage **164** is separate from first passage **162** and does not fluidically connect with first passage **162** in fuel injector **148**.

[0072] Primary fuel manifolds **150** fluidically connect first passage **162** of each fuel injector **148** to first assembly inlet **144** and first assembly outlet **146**. First assembly inlet **144** is fluidically connected to pump outlet **88** of fuel circulator pump **72** by fuel supply line **76**. First assembly outlet **146** is fluidically connected to fuel tank **82** by fuel return line **78**. Pump inlet **90** of fuel circulator pump **72** is fluidically connected to fuel tank **82**. Together, fuel circulator pump **72**, fuel supply line **76**, primary fuel manifolds **150**, first passages **162** of fuel injectors **148**, fuel return line **78**, and fuel tank **82** form a cooling circuit that circulates fuel to cool fuel injectors **148** while gas turbine engine **20** is off and still hot.

[0073] Secondary fuel manifolds **154** fluidically connect second passage **164** of each fuel injector **148** to second assembly inlet **152**. Second assembly inlet **152** is fluidically connected to an outlet of main fuel pump **80** by main pump line **84**. An inlet of main fuel pump **80** is fluidically connected to fuel tank **82**. Together, fuel tank **82**, main fuel pump **80**, main pump line **84**, secondary fuel manifolds **154**, and second passages **164** of each fuel injector **148** form a fuel delivery path that delivers fuel to nozzle exits **160** and combustor **56** when gas turbine engine **20** is on. By having a cooling circuit that is separate from the fuel delivery path, internal valves are not needed in fuel injectors **148**. Furthermore, the cooling circuit can operate simultaneous to the fuel delivery path in the event fuel injectors **148** require additional cooling while gas turbine engine **20** is on.

Discussion of Possible Embodiments

[0074] The following are non-exclusive descriptions of possible embodiments of the present invention.

[0075] A fuel system for a gas turbine engine includes a manifold assembly with an assembly inlet, an assembly outlet, and a plurality of fuel injectors. The manifold assembly also includes at least one fuel manifold fluidically connecting each fuel injector of the plurality of fuel injectors to the assembly inlet and to the assembly outlet. A fuel return line is fluidically connected to the assembly outlet. A fuel circulator pump includes a pump outlet fluidically connected to the assembly inlet.

[0076] The fuel system of the preceding paragraph can optionally include, additionally and/or alternatively, any one or more of the following features, configurations and/or additional

components in the paragraphs below.

[0077] In an embodiment of the foregoing fuel system, the fuel circulator pump further comprises a pump inlet, wherein the pump inlet is fluidically connected to the fuel return line.

[0078] In an embodiment of the foregoing fuel system, the fuel system further comprises: a fuel tank; a main fuel pump comprising: a main pump inlet fluidically connected to the fuel tank; and a main pump outlet fluidically connected to the assembly inlet; and a check valve fluidically connecting the main pump outlet to the assembly inlet.

[0079] In an embodiment of the foregoing fuel system, the fuel circulator pump comprises a shut off valve configured to open and close the pump inlet and/or the pump outlet of the fuel circulator pump.

[0080] In an embodiment of the foregoing fuel system, each fuel injector of the plurality of fuel injectors comprises: a strut; a nozzle head; a supply passage in the strut that is fluidically connected to the at least one fuel manifold; and an internal valve fluidically connecting the supply passage to the nozzle head.

[0081] In an embodiment of the foregoing fuel system, the internal valve is a pressure relief valve.

[0082] In an embodiment of the foregoing fuel system, the internal valve comprises an electrical actuator.

[0083] In an embodiment of the foregoing fuel system, the fuel circulator pump is mechanically coupled to a core turning motor of a gas turbine engine.

[0084] In an embodiment of the foregoing fuel system, the fuel circulator pump is an electrically powered fuel pump.

[0085] In an embodiment of the foregoing fuel system, the fuel system further comprises a fuel tank; a main fuel pump comprising: a main pump inlet fluidically connected to the fuel tank; and a main pump outlet fluidically connected to the assembly inlet; and a check valve fluidically connecting the main pump outlet to the assembly inlet, and wherein the fuel return line is connected to the fuel tank.

[0086] In an embodiment of the foregoing fuel system, the fuel circulator pump further comprises a pump inlet, wherein the pump inlet is fluidically connected to the fuel tank.

[0087] In an embodiment of the foregoing fuel system, the fuel system further comprises: a fuel tank; a second manifold assembly comprising: a second assembly inlet; the plurality of fuel injectors; and at least one secondary manifold fluidically connecting each fuel injector of the plurality of fuel injectors to the second assembly inlet; and a main fuel pump comprising: a main pump inlet fluidically connected to the fuel tank; and a main pump outlet fluidically connected to the second assembly inlet, and wherein a pump inlet of the fuel circulator pump is fluidically connected to the fuel tank and the fuel return line is fluidically connected to the fuel tank.

[0088] In an embodiment of the foregoing fuel system, each fuel injector of the plurality of fuel injectors comprises: a strut; a nozzle head; a first passage in the strut that is fluidically connected to the at least one fuel manifold and fluidically isolated from an exit of the nozzle head; and a second passage in the strut that is fluidically connected to the at least one secondary manifold and to the exit of the nozzle head.

[0089] A method is disclosed for cooling a fuel system of a gas turbine engine. The method includes shutting down the gas turbine engine. After shutting down the gas turbine engine, a fuel circulator pump is activated to flow fuel into at least one fuel manifold and a plurality of fuel injectors fluidically connected to the at least one fuel manifold. The fuel is directed from the at least one fuel manifold and the plurality of fuel injectors into a fuel return line.

[0090] The method of the preceding paragraph can optionally include, additionally and/or alternatively, any one or more of the following features, configurations and/or additional components in the paragraphs below.

[0091] In an embodiment of the foregoing method, the method further includes directing the fuel in the fuel return line to an inlet of the fuel circulator pump.

[0092] In an embodiment of the foregoing method, the method further includes directing the fuel in the fuel return line to a fuel tank, and wherein an inlet of the fuel circulator pump is fluidically connected to the fuel tank.

[0093] In an embodiment of the foregoing method, the method further includes: closing a nozzle exit on each fuel injector of the plurality of fuel injectors after shutting down the gas turbine engine to prevent the fuel from passing through the nozzle exit of each fuel injector of the plurality of fuel injectors; and flowing the fuel radially inward inside each fuel injector of the plurality of fuel injectors relative to a center axis of the gas turbine engine; flowing the fuel radially outward inside of each fuel injector of the plurality of fuel injectors relative to the center axis of the gas turbine engine; and directing the fuel out of each fuel injector of the plurality of fuel injectors and toward the fuel return line.

[0094] In an embodiment of the foregoing method, the method further includes fluidically disconnecting a main fuel pump as part of shutting down the gas turbine engine by closing a check valve that is fluidically connected between an outlet of the main fuel pump and the at least one fuel manifold.

[0095] A fuel injector for a gas turbine engine includes a strut and a nozzle head connected to the strut. The nozzle head includes a nozzle exit. A supply passage extends through the strut to the nozzle head. A return passage extends through the strut toward the nozzle head. A valve is within the strut and/or nozzle head and fluidically connects the supply passage to the nozzle exit.

[0096] The fuel injector of the preceding paragraph can optionally include, additionally and/or alternatively, any one or more of the following features, configurations and/or additional components in the paragraphs below.

[0097] In an embodiment of the foregoing fuel injector, the valve is at least one of a pressure relief valve and an electrically actuated valve.

[0098] While the invention has been described with reference to an exemplary embodiment(s), it will be understood by those skilled in the art that various changes may be made and equivalents may be substituted for elements thereof without departing from the scope of the invention. In addition, many modifications may be made to adapt a particular situation or material to the teachings of the invention without departing from the essential scope thereof. Therefore, it is intended that the invention not be limited to the particular embodiment(s) disclosed, but that the invention will include all embodiments falling within the scope of the appended claims.

Claims

1. A fuel system for a gas turbine engine comprises: a manifold assembly comprising: an assembly inlet; an assembly outlet; a plurality of fuel injectors, each of the plurality of fuel injectors further comprising: a strut extending radially inward from an outer end toward a center axis of the gas turbine engine; a nozzle head connected to the strut opposite the outer end of the strut; a first valve seat in the strut and closer to the nozzle head than to the outer end of the strut, or the first valve seat is in the nozzle head; a second valve seat in the strut and closer to the nozzle head than to the outer end of the strut, or the second valve seat is in the nozzle head; a return passage extending from the outer end of the strut to the first valve seat; a supply passage extending from the outer end of the strut to the second valve seat; a nozzle exit formed on the nozzle head and fluidically connected to the second valve seat; and a valve head between the first valve seat and the second valve seat, wherein in a first position the valve head is against the first valve seat and fluidically opens the nozzle exit to the supply passage and fluidically closing the return passage to both the nozzle exit and the supply passage when the gas turbine engine is on, and wherein in a second position the valve head is against the second valve seat and fluidically closes the nozzle exit to both the supply passage and the return passage and fluidically opens the return passage to the supply passage when the gas turbine engine is off; and at least one fuel manifold fluidically connecting each fuel injector

of the plurality of fuel injectors to the assembly inlet and to the assembly outlet; a fuel return line fluidically connected to the assembly outlet; and a fuel circulator pump comprising: a pump outlet fluidically connected to the assembly inlet.

2. The fuel system of claim 1, wherein the fuel circulator pump further comprises: a pump inlet, wherein the pump inlet is fluidically connected to the fuel return line.

3. The fuel system of claim 2, further comprising: a fuel tank; a main fuel pump comprising: a main pump inlet fluidically connected to the fuel tank; and a main pump outlet fluidically connected to the assembly inlet; and a check valve fluidically connecting the main pump outlet to the assembly inlet.

4. The fuel system of claim 3, wherein the fuel circulator pump comprises: a shut off valve configured to open and close the pump inlet and/or the pump outlet of the fuel circulator pump.

5-6. (canceled)

7. The fuel system of claim 1, further comprising: an electrical actuator; and a drive shaft extending through the strut to connect the valve head to the electrical actuator.

8. The fuel system of claim 1, wherein the fuel circulator pump is mechanically coupled to a core turning motor of a gas turbine engine.

9. The fuel system of claim 1, wherein the fuel circulator pump is an electrically powered fuel pump.

10. The fuel system of claim 1, further comprising: a fuel tank; a main fuel pump comprising: a main pump inlet fluidically connected to the fuel tank; and a main pump outlet fluidically connected to the assembly inlet; and a check valve fluidically connecting the main pump outlet to the assembly inlet, and wherein the fuel return line is connected to the fuel tank.

11. The fuel system of claim 10, wherein the fuel circulator pump further comprises: a pump inlet, wherein the pump inlet is fluidically connected to the fuel tank.

12-20. (canceled)

21. A fuel system for a gas turbine engine comprises: a fuel tank; a first manifold assembly further comprising: a first assembly inlet; a first assembly outlet; a plurality of fuel injectors; and a first fuel manifold fluidically connecting each fuel injector of the plurality of fuel injectors to the first assembly inlet and to the first assembly outlet; a fuel return line fluidically connected to the first assembly outlet; a fuel circulator pump comprising: a circulator pump inlet; a circulator pump outlet fluidically connected to the first assembly inlet; a second manifold assembly comprising: a second assembly inlet; the plurality of fuel injectors; and a second manifold fluidically connecting each fuel injector of the plurality of fuel injectors to the second assembly inlet; and a main fuel pump comprising: a main pump inlet fluidically connected to the fuel tank; and a main pump outlet fluidically connected to the second assembly inlet.

22. The fuel system of claim 21, wherein each fuel injector of the plurality of fuel injectors comprises: a strut; a nozzle head; a first passage in the strut that is fluidically connected to the first fuel manifold and fluidically isolated from an exit of the nozzle head; and a second passage in the strut fluidically connecting the second manifold to the exit of the nozzle head.

23. A gas turbine engine, the gas turbine engine comprising: a low-speed spool; a high-speed spool; a manifold assembly comprising: an assembly inlet; an assembly outlet; a plurality of fuel injectors; at least one fuel manifold fluidically connecting each fuel injector of the plurality of fuel injectors to the assembly inlet and to the assembly outlet; a main fuel pump fluidically connected to the assembly inlet; a fuel circulator pump comprising: a pump outlet fluidically connected to the assembly inlet; a pump inlet fluidically connected to the assembly outlet; a core turning motor mechanically coupled to the fuel circulator pump; and a coupling, wherein the coupling mechanically connects the core turning motor to the low-speed spool.

22. The fuel system of claim 23, wherein the coupling mechanically disconnects the core turning motor to the low-speed spool and the coupling mechanically connects the core turning motor to the high-speed spool.

